Jitter Detection and Image Restoration Based on Continue Dynamic Shooting Model for High-Resolution TDI CCD Satellite Images

Jun Pan, Guo Ye, Ying Zhu, Xiaolin Song, Fen Hu, Chi Zhang, and Mi Wang, *Member, IEEE*

***Abstract*— Although time delay integration charge-coupled devices (TDI CCDs) have been widely used in high-resolution spaceborne optical cameras, they are sensitive to satellite jitter: the images obtained by them are affected by both distortion and blur. Therefore, according to the multistage integral imaging characteristics of TDI CCDs, this article not only proposes a continue dynamic shooting model (CDSM) to reflect the real push-broom mode of the satellite but also presents a method containing jitter detection and image restoration based on it. In the presented method, the CDSM subdivides the TDI CCD integration intervals. The subdivision number of CDSM is determined by the proposed integral transformation function (ITF). Then, it feeds back into the ITF and also contributes to the point spread function (PSF) estimation. Among the abovementioned, ITF defines the relationship between the parallax images and the jitter curve, and aims to improve the jitter detection performance. Finally, an adaptive image restoration based on context is conducted, which combines time, space, and spectrum information. Besides the simulated images, multispectral images of GaoFen-1 02 satellite were also adopted to validate the performance of the presented method. Experimental results indicate that the accuracy of the jitter detection is increased, and the geometric and radiometric qualities of restored images are also improved.**

***Index Terms*— Distortion and blur, image restoration, jitter detection, point spread function (PSF), satellite jitter, time-delay integration charge-coupled device (TDI CCD).**

I. Introduction

THE time delay integration charge-coupled device (TDI CCD) is a type of linear CCD based on time-delay integration (TDI) technology, developed on the idea of obtaining multiple exposures of the same target to increase the total amount of light captured. Since the TDI CCD can increase the luminous flux, it can also improve the sensitivity and signal-to-noise ratio of a camera. For this reason, it has become the mainstream sensor of choice in high-resolution spaceborne optical imaging systems [1]–[4].

However, the use of integral imaging techniques makes TDI CCDs more vulnerable to satellite jitter [5], [6]. Satellite jitter is an error source caused by satellite attitude adjustment and the periodic movement of satellite parts during orbit operation [7], [8]. Due to the sensitivity and uncontrollability of satellite jitter, its impact is difficult to eliminate through physical means. The offsets are produced when satellite jitter is transmitted to the image plane. They change the geometric and radiometric information of the image and directly lead to a decline in image quality. Moreover, different from ordinary CCDs, jitter offsets with TDI CCDs tend to accumulate during the imaging process, for exacerbating the mismatch between image motion speed and charge transfer rate and aggravating the image degradation degree [3]. Many satellites equipped with TDI CCDs are affected by satellite jitter [9]–[11]. In addition, the impacts of satellite jitter are worsening with the improvement of satellite spatial resolution [12], [13]. Therefore, the exploration of a method to restore high-resolution TDI CCD images affected by satellite jitter has become a vital matter of research.

The premise for the restoration of high-resolution TDI CCD images affected by satellite jitter is its accurate jitter detection through direct or indirect methods. Direct methods mainly rely on the collection of satellite angular displacement, velocity, and acceleration data through highly sensitive angle sensors; unfortunately, there are very few satellites equipped with certain specified high-frequency measuring equipment [10], [14]–[16]. In contrast, indirect methods can be realized in an economical and convenient way by extracting the jitter curve in the field of digital image processing [11], [17]–[20], and there have been many related studies. For example, Teshima and Iwasaki [17] investigated the attitude fluctuation using short-wave infrared (SWIR) images of Terra spacecraft. Amberg *et al.* [39] and Hu *et al.* [40] conducted the high-frequency jitter detection with multispectral images used for PLEIADES-HR satellites and Chinese Mapping Satellite-I, respectively. Tong *et al.* [13] used three-line array stereo images to detect the periodic distortions of ZY-3 that are caused by satellite jitter. Although numerous jitter detection methods are available,

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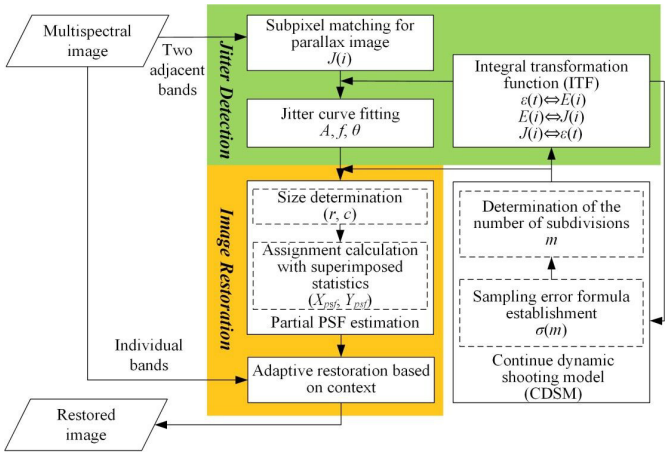

Fig. 1. Workflow of the method presented in this article.

most of them only focus on general push-broom cameras without taking full care of the TDI technique.

Image distortion and blur are coupling effects caused by satellite jitter, but most of the previous studies have tried to solve only one of them at a time. For image distortion, there are two typical ways to compensate. One involves the direct image distortion correction through a resampling that takes into account the registration error; the other involves the satellite attitude correction taking into account the offset of the attitude angles [8], [17], [18], [21]–[23]. For image blur, the solutions can be divided into blind and nonblind restoration methods. Blind restoration is usually used when the blur kernel is unknown, and the typical algorithms are the maximum likelihood deconvolution method, the iterative blind deconvolution (IBD) method, blind deconvolution method by means of the Richardson–Lucy algorithm, and so on [42]–[45]. If the point spread function (PSF) is explicitly known prior to the deconvolution, nonblind restoration methods, such as Wiener filtering, constrained least squares filtering, and inverse filtering, can be adopted. Among most studies about satellite jitter, the same PSF is usually applied to the entire image, due to the consideration that the motion of jitter can be regarded as uniform and linear within a short imaging time [24]–[26]. However, this assumption is valid only for low-frequency jitter, because the complex temporal changes appear for high-frequency jitter.

Since geometric and radiometric information are highly coupled, they should be considered simultaneously, rather than separately. Therefore, this article proposes a method containing jitter detection and image restoration for high-resolution TDI CCD images affected by satellite jitter. The innovation mainly lies in the following: 1) the proposed integral transformation function (ITF) increases the jitter detection accuracy because it can decompose the overall effect in immediate effects and extract the immediate jitter offsets in the object space; 2) the presented continue dynamic shooting model (CDSM) takes fully into account the jitter temporal variation and the continuous push-broom mode of the TDI CCD, and makes the adaptive restoration based on context reduce the coupled distortion and blur effects simultaneously.

II. Methodology

In this article, a method containing jitter detection and image restoration based on CDSM is proposed to improve the jitter detection performance and eliminate the coupled effect of distortion and blur. The workflow shown in Fig. 1 consists of two stages (jitter detection and image restoration), encompassing the entire process to deal with the degradation of TDI CCD images affected by jitter. These two stages are both based on CDSM, which is presented in this article to reflect the real push-broom mode of the satellite by a denser sampling strategy. The first stage involves the calculation of the parallax image by subpixel matching, and the construction of ITF according to the imaging principle of the TDI CCD, where subpixel matching is achieved by correlation coefficient and least squares algorithm [11]. The ITF provides the guidance for the determination of subdivisions in CDSM and then the specific CDSM is fed back into ITF to make a more precise jitter curve fitting. In the second stage, the correspondent partial PSF is estimated accurately with the assistance of CDSM, and finally, an adaptive restoration based on context completes the whole process. The main steps of the process are described in the following sections.

A. Subpixel Matching

A lot of points matched on scanning lines are essential for parallax images, and point-by-point dense matching strategy between adjacent bands is demanded. Hence, a subpixel matching method relying on the correlation coefficient and the least squares algorithm [11] is adopted in the stage of jitter detection.

The subpixel matching is achieved through the two-step way, containing coarse matching followed by fine matching. The detailed process is described as follows. First, the correlation matching is utilized to get the coarse corresponding points. Due to the settled coordinate difference between bands on the camera focal, points on each two fixed apart lines in adjacent bands are taken as the target object. And those eight-neighborhood points around the initial one are also taken as the candidates. The correlation coefficients between matched point pairs are calculated, and then the candidate with the maximum correlation is picked as the coarse corresponding point. Second, with the coarse corresponding point as the center of the matching window, the transformation parameters are obtained by the iterative least squares solution, so as to attain the refined corresponding point. In the iterative process of the least square matching, a certain threshold of correlation coefficient and a maximum of iteration times are used to filter out the mismatching points. In this way, the correlation matching provides the whole-pixel corresponding point, and the least square matching further achieves a subpixel accuracy. As shown in the study by Gruen [46], the matching precision of this method can reach up to 1/10 pixel and even better.

B. ITF

For better detection of satellite jitter, ITF is proposed to describe the relationship between the parallax image and the

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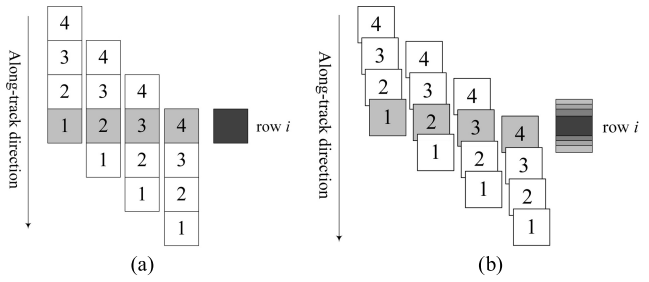

Fig. 2. Integral imaging process of the TDI CCD. (a) Ideal situation without jitter. (b) Actual situation with jitter.

jitter curve. The images produced by TDI CCD are affected by the accumulation of jitter offsets during multistage integration imaging. A detailed analysis requires the extraction of jitter offsets corresponding to specific moments; to achieve this objective, the overall effect needs to be decomposed into immediate effects. The ITF allows the conversion of the parallax image [$J(i)$, where i stands for the line number of the image] into a jitter curve [$\varepsilon(t)$, where t stands for the time]. This conversion is in fact a transformation from the image space to the real object space, because $J(i)$ reflects the image motion change with rows and $\varepsilon(t)$ reflects the relationship between the time-varying jitter and time.

The TDI CCD is an area array CCD with a linear array working mode. Its most distinctive characteristic is the multistage integral imaging process. A multirow linear array CCD exposes the target in turn and records it on different rows with the photoinduced charge bag transfer and accumulation [27]. Fig. 2(a) displays the ideal imaging process of row i, considering a TDI CCD integration stage of 4. In this case, the exposure of each level of integration corresponds to the same ground area. However, the offsets caused by satellite jitter lead to the successive dislocation and aliasing of charges in different regions, resulting in image distortion and blur, as shown in Fig. 2(b). The working principles of the TDI CCD amplify the effects of satellite jitter.

During the TDI CCD conducting the imaging process, jitter offsets are continuously accumulated and influence the location of image ground features. The ITF describes the relationship between changes in the jitter offsets related to time and to different rows. To simplify the analysis, the TDI CCD exposure is usually regarded as a discrete operation: the addition of multiple jitter offsets is performed considering discrete points along the jitter curve [1]–[3]. The imaging time of each row is divided into N equal parts, where N corresponds to the TDI CCD integration stages, and the charge bags vary at the endpoint of each interval. Take the row i for instance, the integral result can be expressed as follows:

$$E(i) = \frac{1}{N}\left[\frac{1}{2}\varepsilon((i-N)T) + \sum_{k=1}^{N-1}\varepsilon((i-N+k)T) + \frac{1}{2}\varepsilon(iT)\right] \tag{1}$$

where $E(i)$ represents the image motion caused by jitter in row i, $\varepsilon(t)$ is the jitter curve (i.e., the relationship between jitter offsets and time), and T is the time needed for one

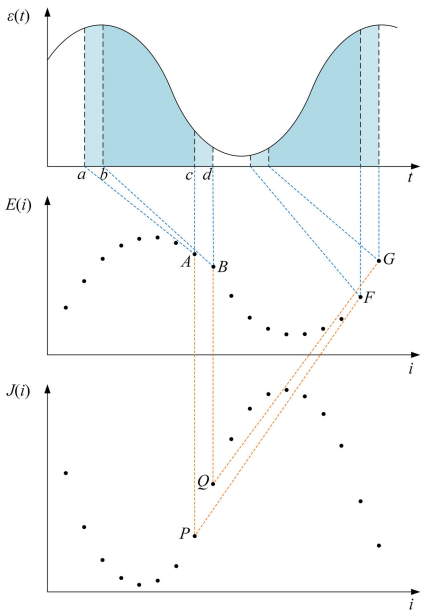

Fig. 3. $\varepsilon(t)$-to-$J(i)$ evolving process as described by the ITF.

integration. Through subpixel matching based on the correlation coefficient and least squares algorithm [11], the relative registration error between two adjacent bands of the multispectral degraded image can be calculated, and then the parallax image [$J(i)$] is obtained. The purpose of the ITF is to describe the relationship between $\varepsilon(t)$ and $J(i)$. Multispectral bands are generally placed parallel to the focal plane; hence, the coordinate difference between corresponding points on two different bands of the multispectral image is almost fixed [i.e., $J(i) = E(i+L) - E(i)$, where L is the distance between the two bands]. Thus, the ITF is defined as follows:

$$J(i) = \frac{1}{N}\left\{\frac{1}{2}[\varepsilon((i+L-N)T) + \varepsilon((i+L)T) - \varepsilon((i-N)T) - \varepsilon((i)T)] + \sum_{k=1}^{N-1}[\varepsilon((i+L-N+k)T) - \varepsilon((i-N+k)T)]\right\}. \tag{2}$$

The ITF formulates the $\varepsilon(t)$-to-$J(i)$ transition with $E(i)$ as an intermediary. Fig. 3 shows the specific evolving process. Point A in the image motion $E(i)$ can be obtained from the integration of the offsets comprised in the interval (a, c) below the jitter curve $\varepsilon(t)$, while point B on the next row can be obtained from the integration of the offsets comprised in the interval (b, d). In the above-described case, (a, c) and (b, d) correspond to the single-row imaging time; In addition, (a, b) corresponds to the single integration time of the TDI CCD. Points F and G in $E(i)$ can be obtained in the same way as points A and B. Point P in the parallax image $J(i)$ can be calculated from the image motion differences between corresponding rows on adjacent bands [it corresponds to points A and F in $E(i)$]. Similarly, point Q on the next row can be obtained by points B and G in the same way as point P.

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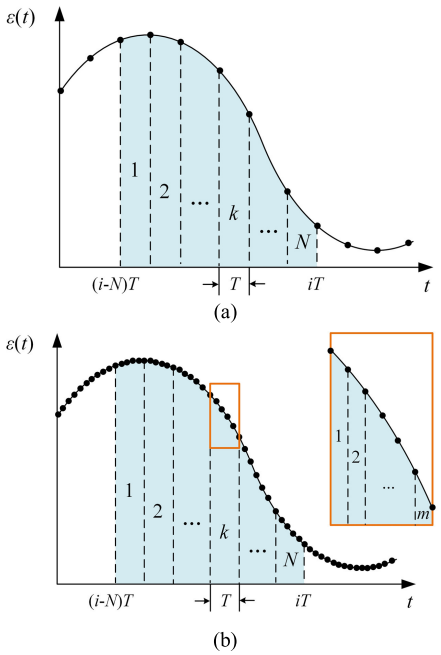

Fig. 4. Segmentation of imaging time and selection of sampling points. (a) Illustration of conventional methods. (b) Illustration of CDSM.

C. CDSM

The CDSM is presented to provide a denser sampling strategy to reflect the continuous push-broom mode of cameras. The CDSM can make it more reasonable to deal with continuous imaging process in the discrete way and reduce the associated error.

1) Fundamentals of CDSM: Remote sensing images are acquired through the continuous push-broom process of satellites. The TDI CCD scans one pixel at once integral, and its integration time is equal to the exposure time of the camera [29], [30]. However, since the single integration time of a TDI CCD is short, the correspondent photoelectric conversion and accumulation of exposure energy are usually regarded as instantaneous [1]–[3]. Sampling is typically done at the end of each time interval, as shown in Fig. 4(a) [where $\varepsilon(t)$ is the jitter curve, and N is the number of integration stages of the TDI CCD]. A high ratio between the imaging time and the jitter curve period indicates a high variability of offsets. In this case, the common discretization pattern might ignore the jitter offsets with complex changes hidden in the interval, which also contribute to the image degradation. Thus, the CDSM is proposed. Its main characteristic is the high number of sampling points used for the discretization of the TDI CCD exposure process. Through such way, the CDSM provides a good representation of the actual push-broom process of satellites by modeling that cameras are in a continuous dynamic shooting mode. Fig. 4 indicates the differences between conventional methods and CDSM when determining the segmentation of imaging time and selecting sampling points (the imaging of row i is taken as an example displayed in blue). Fig. 4(a) shows the conventional segmentation and sampling points: the single-row imaging time is divided into N equal parts and the sampling points correspond to the endpoints of each of these parts. Fig. 4(b) takes interval k, for instance, and the interval is marked with orange box, showing that the CDSM would further subdivide each interval into m equal parts and select sampling points at the end of these parts. By applying the CDSM, the single-row imaging time would be segmented into a total of $N*m$ parts, obtaining a total of $(N*m+1)$ sampling points.

2) Sampling Error Formula Establishment: The discrete processing of the TDI CCD photoelectric transfer results in an inevitable error between the results of the theoretical calculation and the real continuous case. The extent to which the CDSM can reduce this error depends on the number of subdivisions (i.e., m). The sampling error formula is established according to the ITF to explore the relationship between m and the corresponding error.

The jitter curve can be described by the sine function, i.e., $\varepsilon(t) = A\sin(2\pi t/T_j + \theta)$, where A is the amplitude, T_j is the period of the jitter curve, and θ is the initial phase [31]. The above-described CDSM is developed from the conventional sampling pattern reflected in (1). After it is introduced, the image motion of row i caused by jitter $[E(i)]$ in (1) becomes $E(i,m)$, which is affected by the number of subdivisions in the CDSM (m)

$$E(i,m) = \frac{1}{mN}\left[\frac{1}{2}A\sin\left((i-N)\frac{2\pi T}{T_j}+\theta\right) + \sum_{k=1}^{mN-1} A\sin\left(\left(i-N+\frac{k}{m}\right)\frac{2\pi T}{T_j}+\theta\right) + \frac{1}{2}A\sin\left(i\cdot\frac{2\pi T}{T_j}+\theta\right)\right] \tag{3}$$

where N represents the number of integration stages of the TDI CCD, and T is the single integration time of the TDI CCD. The imaging time of a row is denoted by T_r, and then $T_r = NT$.

The imaging process is the same for each line of the image. To simplify the analysis, the row, whose imaging time is comprised in the interval $(0, T_r)$, is adopted for an example. The image motion caused by jitter of that row is noted as $S(m)$, and it can be expressed by the following equation:

$$\begin{aligned} S(m) &= \frac{1}{mN}\left[\frac{1}{2}A\sin\theta + \sum_{k=1}^{mN-1} A\sin\left(\frac{2\pi}{T_j}\cdot\frac{kT_r}{mN}+\theta\right) + \frac{1}{2}A\sin\left(\frac{2\pi}{T_j}\cdot T_r+\theta\right)\right] \\ &= \frac{1}{mN}\left[\frac{1}{2}A\sin\theta + A\frac{\cos\left(\frac{\pi}{mN}\cdot\frac{T_r}{T_j}+\theta\right)-\cos\left(\left(2-\frac{1}{mN}\right)\cdot\frac{\pi T_r}{T_j}+\theta\right)}{2\sin\left(\frac{\pi}{mN}\cdot\frac{T_r}{T_j}\right)} + \frac{1}{2}A\sin\left(\frac{2\pi}{T_j}\cdot T_r+\theta\right)\right]. \end{aligned} \tag{4}$$

Equation (4) displays the jitter output, considering the imaging process of TDI CCD in a discrete way. However, the true value

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of the jitter offset (C) following a continuous camera push-broom is:

$$C = \frac{1}{T_r}\int_0^{T_r} A\sin\left(\frac{2\pi}{T_j}t+\theta\right)dt = \frac{A}{2\pi}\cdot\frac{T_j}{T_r}\left[\cos\theta-\cos\left(2\pi\cdot\frac{T_r}{T_j}+\theta\right)\right]. \quad (5)$$

The relative error between the true value (C) and the calculated value [$S(m)$] can reflect the error caused by the sampling density due to the number of subdivisions in the CDSM (m). Therefore, the sampling error formula is established as follows:

$$\begin{aligned}\sigma(m) &= \left|\frac{S(m)-C}{C}\right| \\ &= \left|\frac{1}{mN}\left\{\sin\theta+\cot\left(\frac{\pi}{mN}\cdot\frac{T_r}{T_j}\right)\left[\cos\theta-\cos\left(\frac{2\pi T_r}{T_j}+\theta\right)\right]\right.\right. \\ &\quad \left.\left.-\left[\sin\theta+\sin\left(\frac{2\pi T_r}{T_j}+\theta\right)\right]+\sin\left(\frac{2\pi}{T_j}\cdot T_r+\theta\right)\right\}\right. \\ &\quad \left.-\frac{1}{\pi}\cdot\frac{T_j}{T_r}\left[\cos\theta-\cos\left(2\pi\cdot\frac{T_r}{T_j}+\theta\right)\right]\right| \Big/ \\ &\quad \left|\frac{1}{\pi}\cdot\frac{T_j}{T_r}\left[\cos\theta-\cos\left(2\pi\cdot\frac{T_r}{T_j}+\theta\right)\right]\right| \\ &= \left|\frac{\pi}{mN}\cdot\frac{T_r}{T_j}\cot\left(\frac{\pi}{mN}\cdot\frac{T_r}{T_j}\right)-1\right|. \end{aligned} \quad (6)$$

3) Determination of the Number of Subdivisions: The precondition, the number of subdivisions in the CDSM (m) should satisfy, is Shannon's sampling theorem to avoid aliasing effect. Moreover, when determining the value m, we should not only take into account reducing the sampling error but also minimize the computational complexity. The theoretical analysis is described here, and the experimental verification is conducted in Section III-A.

Equation (6) indicates that the sampling error ($\sigma(m)$) is directly related to m and affected by the ratio between T_r and T_j, as well as by the number of integration stages (N). For a certain jitter, the deviation of the fitted curve is positively related to the imaging time; thus, the error tends to increase with $\frac{T_r}{T_j}$. Fig. 5(a) shows the relationship between the sampling error [$\sigma(m)$] and the number of subdivisions (m) for different values of the time ratio (T_r/T_j), in the case of $N=4$. The number of integration stages (N) is an inherent attribute of the TDI CCD and varies for each camera. Adopting a TDI CCD with a higher N has the same effect of increasing the m value in the CDSM: the sampling density increases and the error is reduced [see Fig. 5(b), where the time ratio is set at 2.0]. Nevertheless, independently from T_r/T_j and N, larger m values in the CDSM will correspond to smaller errors. In addition, these errors tend to remain stable despite continuous increases in m. Fig. 5 shows how the error remains relatively stable for $m \geq 6$; therefore, this article considers 6 as a reasonable number of subdivisions.

After the number of subdivisions (m) in the CDSM is determined, (3) is used instead of (1), then (2) which formulates the ITF is transformed into

$$\begin{aligned} J(i) &= \frac{A}{mN}\left\{\frac{1}{2}\left[\sin\left((i+L-N)\frac{2\pi T}{T_j}+\theta\right)+\sin\left((i+L)\frac{2\pi T}{T_j}+\theta\right)\right]\right. \\ &\quad -\frac{1}{2}\left[\sin\left((i-N)\frac{2\pi T}{T_j}+\theta\right)-\sin\left(i\cdot\frac{2\pi T}{T_j}+\theta\right)\right] \\ &\quad +\sum_{k=1}^{mN-1}\left[\sin\left(\left(i+L-N+\frac{k}{m}\right)\frac{2\pi T}{T_j}+\theta\right)\right. \\ &\quad \left.\left.-\sin\left(\left(i-N+\frac{k}{m}\right)\frac{2\pi T}{T_j}+\theta\right)\right]\right\}. \end{aligned} \quad (7)$$

As thus, the CDSM specialized by the ITF is reapplied to the ITF. The jitter curve hence has a more detailed description, so that the jitter detection will be more accurate.

D. Partial PSF Estimation

Due to the jitter temporal variation, the PSF tends to vary from one row to the other. A more detailed partial PSF can be estimated through the CDSM.

The PSF of a moving degraded image can be represented by a probability density function (PDF), which includes the relative position of the ground object image with respect to the detector during the imaging process [32]. The PSF needed for image restoration was derived from the analysis of a line spread function (LSF). For TDI CCD images, the LSF for row i [LSF(i)] during the single integration time (T) is the normalized energy distribution

$$\mathrm{LSF}(i) = \frac{1}{T}\int_{t_0}^{t_0+T} w(i,t)dt \quad (8)$$

where t_0 is the initial moment of the imaging process, and $w(i,t)$ is the instantaneous energy distribution at moment t in row i. Since $w(i,t)=\delta(i-\varepsilon(t))$, (8) can be written as

$$\mathrm{LSF}(i) = \frac{1}{T}\int_{t_0}^{t_0+T} \delta(i-\varepsilon(t))dt \quad (9)$$

where $\delta(\cdot)$ is the impulse function, and $\varepsilon(t)$ is the jitter offset corresponding to the along-track direction [$X(t)$] and across-track direction [$Y(t)$]. The PSF of row i [PSF(i)] can be calculated as the multiple integral of LSF(i)

$$\mathrm{PSF}(i) = \frac{1}{NT}\sum_{k=1}^{N}\int_{(i-N+k-1)T}^{(i-N+k)T}\delta(i-X(t),i-Y(t))dt \quad (10)$$

where N represents the number of integration stages of the TDI CCD. Taking the discrete operation for the TDI CCD exposure time via the CDSM, which is described in Section II-B, (10) can be expressed as

$$\begin{aligned}\mathrm{PSF}(i) = \frac{1}{mN}\left[\frac{1}{2}\delta(i-X((i-N)T),i-Y((i-N)T))\right. \\ +\frac{1}{2}\delta(i-X(iT),i-Y(iT)) \\ +\sum_{j=1}^{mN-1}\delta\left(i-X\left(\left(i-N+\frac{j}{m}\right)T\right),\right. \\ \left.\left.i-Y\left(\left(i-N+\frac{j}{m}\right)T\right)\right)\right]\end{aligned} \quad (11)$$

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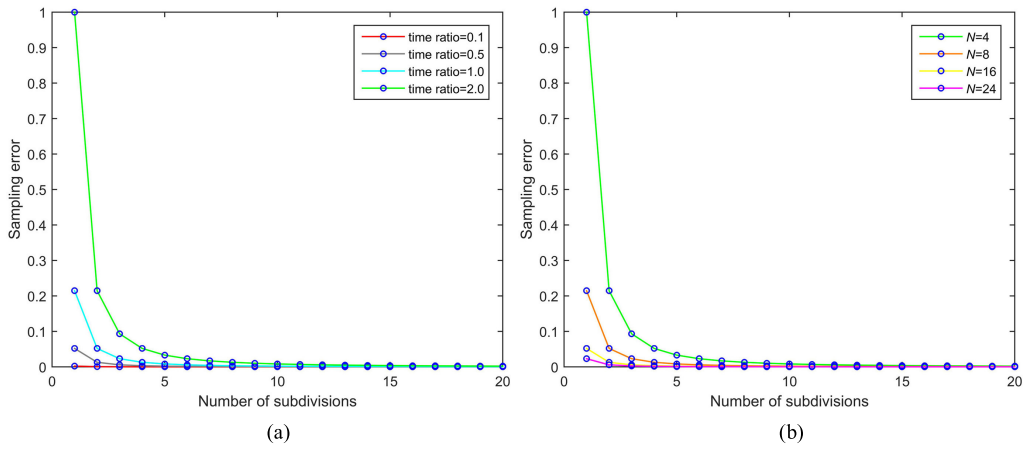

Fig. 5. Relationship between sampling error and number of subdivisions in CDSM. (a) Considering different time ratio values. (b) Considering different integration stages of TDI CCD.

where m is the number of subdivisions in the CDSM. According to (11), the PSF is space-variant due to the jitter temporal variation: the PSF of each row has to be calculated separately. The concrete estimation of the PSF is conducted through two main steps: size determination and assignment calculation with superimposed statistics.

1) Size Determination: The PSF size varies for different image rows and is determined by the maximum value reached by the jitter curve during the imaging period of that specific row. For example, in the generation process of a row, a indicates the maximum value of the jitter offset in the along-track direction, while b indicates that in the across-track direction. In order to reduce the computational complexity and avoid unnecessary operations in the subsequent restoration, because the PSF supposes an odd number of rows and columns, the size of the PSF for the row is defined as

$$\begin{pmatrix} r \\ c \end{pmatrix} = 2\begin{pmatrix} \lceil |a| \rceil \\ \lceil |b| \rceil \end{pmatrix} + 1 \tag{12}$$

where r and c represent the number of rows and columns, respectively; the $\lceil \cdot \rceil$ operation selects the minimum integer greater than or equal to the value inside. The initialization is performed on the empty matrix PSF after size determination.

2) Assignment Calculation With Superimposed Statistics: The value of the PSF is directly related to the relative offset between the ideal and the actual imaging center $((\Delta X, \Delta Y))$ at each exposure moment. The ideal imaging center is defined as the geometric center of the PSF, while the location of the actual imaging center $[(X_{\text{psf}}, Y_{\text{psf}})]$ at each discrete moment is calculated as follows:

$$\begin{pmatrix} X_{\text{psf}} \\ Y_{\text{psf}} \end{pmatrix} = \begin{pmatrix} \left\lceil \frac{r}{2} \right\rceil + \Delta X \\ \left\lceil \frac{c}{2} \right\rceil + \Delta Y \end{pmatrix}. \tag{13}$$

The resampling is performed to obtain the weights of the related points in the PSF when $(X_{\text{psf}}, Y_{\text{psf}})$ corresponds to noninteger values. This way is equivalent to the backward induction of bilinear interpolations. We can indicate the adjacent pixels surrounding $(X_{\text{psf}}, Y_{\text{psf}})$ with (p, q), $(p, q+1)$, $(p+1, q+1)$, and $(p+1, q)$; moreover, the distance between (p, q) and $(X_{\text{psf}}, Y_{\text{psf}})$ in two directions can be indicated with u and v. In this case, $(1-u)(1-v)$, $(1-u)v$, uv, and $u(1-v)$ will represent the weights of the four nearest pixels. This kind of calculation is performed at all discrete moments included in the CDSM, covering the whole single-row imaging period. Therefore, for any point of the PSF, multiple weights are calculated at different times; the sum of all calculated weights is assigned to the point.

E. Adaptive Restoration Based on Context

The partial PSF calculated in Section II-C combines the spectral and temporal dimensions. Moreover, the adaptive restoration based on context takes into account the spatial information: the uniqueness of each row, and the similarities between neighboring rows. This process is carried out through optimal window Wiener filtering, by means of multiple inputs and a single output.

The time-varying jitter makes each row of TDI CCD images degenerate to different degrees and the PSF of each row is space-variant. Therefore, images should be restored row by row. Meanwhile, adjacent rows have structural similarities: the jitter curve segments corresponding to overlapping of adjacent rows considerably due to the imaging principle of the TDI CCD (shown in the jitter curve $[\varepsilon(t)]$ in Fig. 3). Therefore, it is appropriate to involve context information in the restoration of a single row, and hence an adaptive restoration approach based on context is adopted. Specifically, the adaptive restoration based on context is conducted in a multiple-input single-output mode: the row to be restored and its neighboring rows are grouped in an image block as the input; then, the middle row of this restored block is taken as the output. That is, the image is restored line by line, but in a way by processing the image block. For row i, the size of its corresponding image block to be restored depends on its PSF size. If the PSF size is indicated as $r * c$ [as shown in (12)], the image block to be restored is chosen at least in the rows comprised between $[i-r, i+r]$. The PSF corresponding to row i is utilized on that image block, and the detailed calculation of the PSF has been demonstrated in Section II-C.

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A method typically applied to image restoration is the Wiener filtering algorithm, which can suppress noise and deal with details. If the nondegraded image is represented by f, and its estimate is represented by $\hat{f}$, the mean square error between them is measured as follows:

$$e^2 = E\left\{\left(f - \hat{f}\right)^2\right\} \tag{14}$$

where $E\{\cdot\}$ is the expectation of the parameters. The Wiener filter is designed to minimize such mean square error, and it can be expressed in the frequency domain as follows [28]:

$$\begin{aligned}\hat{F}(u,v) &= \left[\frac{H^*(u,v)S_f(u,v)}{S_f(u,v)|H(u,v)|^2 + S_\eta(u,v)}\right]G(u,v) \\ &= \left[\frac{H^*(u,v)}{|H(u,v)|^2 + S_\eta(u,v)/S_f(u,v)}\right]G(u,v) \\ &= \left[\frac{1}{H(u,v)}\frac{|H(u,v)|^2}{|H(u,v)|^2 + S_\eta(u,v)/S_f(u,v)}\right]G(u,v)\end{aligned} \tag{15}$$

where $H(u,v)$ is the degradation function in the frequency domain, $H^*(u,v)$ is the complex conjugate of $H(u,v)$, $S_\eta(u,v)$ is the power spectrum of the noise, and $S_f(u,v)$ is the power spectrum of the nondegraded image (f). Equation (15) is based on the following principle: the product of a complex member and its complex conjugate is equal to the square of the amplitude of that complex member, i.e., $H(u,v)H^*(u,v) = |H(u,v)|^2$. If the signal-to-noise ratio is denoted as K, (15) can be written as

$$\hat{F}(u,v) = \left[\frac{1}{H(u,v)}\frac{|H(u,v)|^2}{|H(u,v)|^2 + K}\right]G(u,v). \tag{16}$$

However, the single Wiener filtering algorithm is usually affected by the edge error, which leads to the ringing effect [33]. In this study, we adopted an optimal window Wiener filtering, which can suppress the ringing effect at the edges of the image while maintaining the typical advantages of the original method [34]. The main idea at the base of the optimal window Wiener filtering is the application of a window function to the image before its restoration. The window function is divided into nine regions, whose calculation is based on the amount of motion experienced by each segment, as shown at the bottom of next page. Denoted by $w(x,y)$, the window function for the image block, corresponding to the row to be restored, can be represented as follows: where $h(m,n)$ is the representation of the degradation function in the spatial domain (i.e., the PSF of the image row to be restored); V and H are, respectively, the size of the window function, and they are equal to the size of the image block which needs to restore; r and c are, respectively, the rounding maximum value of the jitter offsets in both directions. Besides this, the sizes of the four regions in the corner of the window function are all $(r-1, c-1)$, where (r,c) is equal to the PSF size. If the degraded image block is $g(x,y)$, the windowed degraded image block can be indicated as $g'(x,y) = g(x,y)w(x,y)$. Since the value of the window function $[w(x,y)]$ is equal to 1 within the central region but gradually approaches 0 toward the image edges, the central area of the windowed image block $[g'(x,y)]$ is equal to that without the window function $[g(x,y)]$, while the value of the edge area gradually tends to 0. In this way, the optimal window Wiener filtering can reduce the gradient jump toward the edge of the image and suppress the ringing effect. It is worth noting that the PSF is space-variant due to the jitter temporal variation, while the window function and degradation function is calculated based on the variable PSF that matches the row to be restored; therefore, the restoration is adaptive.

Fig. 6. Orthoimage for simulation experiments.

III. Experiments and Discussion

We validated the presented method through the simulation and real image experiments. In the simulation experiments, jitters of different frequencies were added to a nondegraded orthoimage; then, the simulated images with jitter were restored applying the presented method. Our method was instead directly applied in the case of the real image experiments.

A. Simulation Experiments

Many studies have shown that jitter offsets in the along-track direction are covered up due to the influence of terrain and other factors, and satellite jitter mainly exits in the across-track direction, so this article conducts simulation experiments in this representative direction [11], [13], [17], [19]. We simulated both low-frequency (time ratio less than 1.0) and high-frequency jitters (time ratio greater than 1.0) [35]. For the simulated low-frequency jitters, the ratio between the single-row imaging time and the jitter period was set to 0.1, while for high-frequency jitters, it was set to 2. For a fixed time ratio and the number of subdivisions in CDSM, a lower number of TDI CCD integration stages correspond to larger errors (see analysis in Section II-B). In order to magnify the error to observe trends in detail, we set the number of integration stages to 4 and considered a single integration time of 2.5 ms. The simulated jitters are expressed as $\varepsilon(t) = 1.5\sin(2\pi f t + 1.5)$ ($f = 10$ Hz corresponding to 0.1 time ratio for low-frequency jitters and $f = 200$ Hz corresponding to 2.0 time ratio for high-frequency jitters). The orthoimage used in the experiments is from SPOT 6 and corresponds to the Beijing area. Its size is 512×512 pixels, and its spatial resolution is 1.5 m (Fig. 6).

A comparison between the method based on the jitter displacement estimation model [20] and the presented method, was conducted to evaluate the jitter detection performance.

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TABLE I
DETECTED PARAMETERS OF THE OBTAINED 0.1 TIME RATIO JITTER CURVE

		Frequency (Hz)	Amplitude (px)	Phase (rad)
The method based on the jitter displacement estimation model	detected	9.9814	1.2676	1.2708
	error	0.0186	0.2324	0.2292
The presented method	detected	9.9814	1.2886	1.5844
	error	0.0186	0.2114	0.0844

Table I shows the parameters of the 0.1 time ratio jitter curve obtained by these two methods. It is proved that the method presented in this article has higher detection accuracy, which is caused due to the ITF and CDSM used in the presented method. The ITF decomposed the overall effect into immediate effects and realized a transformation of jitter offsets from the image space to the real object space, while the CDSM provided a dense sampling strategy that reflected the continue push-broom mode of TDI CCD. In contrast, the method based on the jitter displacement estimation model only paid attention to the overall effect of satellite jitter in the image space. In another word, compared with the method based on the jitter displacement estimation model, the presented method takes fully into account the jitter temporal variation.

Significantly, due to the limitation of subpixel matching accuracy, the parallax image of the 2.0 time ratio jitter cannot be obtained; therefore, the method based on the jitter displacement estimation model and the presented method are no longer suitable. The issue about the image matching for high-frequency jitter will be investigated in our future work. Fig. 7(a)–(d) shows the simulated and corresponding restored images obtained by our method (using the known parameters of simulated jitters in restoration). Those on the left are the images of the experimental results with a local region in the marked rectangle, and details of the local region are shown on the right. Different degrees of distortion and blur are visible in the simulated images. When the jitter frequency is low, the jitter can be taken as linear movement added to the imaging process, so there is mainly distortion in the degraded image [as shown in Fig. 7(a)]; Fig. 7(b) shows the restored image where such distortion is eliminated by the presented method. If the jitter frequency is high, multiple jitter periods will occur during the imaging time, so that the diameter of the blur circle will be expended and blur is significant in the degraded image [as shown in Fig. 7(c)]. In the restored image [as shown in Fig. 7(d)], the blur is reduced. The visualized results prove the effectiveness of the presented method.

In addition, the mean squared error (MSE), quantifying the error between the restored images and the orthoimage, is adopted to evaluate the restoration performance. Fig. 8 shows the relationship between the MSE and the number of subdivisions in the CDSM. The red and green lines highlight the experimental result considering 0.1 time ratio jitter and 2.0 time ratio jitter, respectively. The observed trends are consistent with the theoretical derivation displayed in Fig. 5 (Section II-B): the MSE (the evaluation index used to measure sampling error in Fig. 8) is stable when the number of subdivisions in the CDSM is greater than 6. This result, revealing that 6 is a reasonable number of CDSM subdivisions, confirms the correctness of our analysis. Meanwhile, Fig. 8 shows that when the jitter is in high time ratio, the MSE decreases significantly with the number of subdivisions in the CDSM increasing, while the change is not obvious when the jitter is in low time ratio. This is because the complex jitter temporal variation (the advantage of our method lies in the attention given to it by the presented CDSM) is evident when the jitter is in high time ratio. In general, simulation experiments demonstrate that our method has shown good performance both on jitter detection and image restoration.

B. Real Image Experiments

1) Experimental Data: A number of multispectral images of GaoFen-1 02 satellite have been employed to test the proposed algorithm, and two data sets are presented. GaoFen-1 02 satellite is equipped with 8-m optical cameras, capable of recording images in four spectral bands: blue (B1), green (B2), red (B3), and infrared (B4). The B1, B2, and B3 bands are adopted in this article. The presented two data sets were originally collected on April 15, 2018, and were stored in a 16-bit image format. Image A covers an area near Qingdao, Shandong Province (120.1°E, 36.9°N), while image B covers an area near Huai'an, Jiangsu Province (119.2°E, 33.6°N). Detailed information about the two used images is listed in Table II.

2) Results and Analyses:

a) Results of jitter detection: Fig. 9 shows the parallax images of experimental data after subpixel matching. For image A, the corresponding parallax images obtained by matching B1-B2 and B2-B3 bands are demonstrated in Fig. 9(a) and (b). For image B, the corresponding parallax images obtained by matching B1-B2 and B2-B3 bands are demonstrated in Fig. 9(c) and (d). In addition to the parallax

$$w(x,y)=\begin{bmatrix} \sum_{m=0}^{x}\sum_{n=0}^{y}h(m,n) & \sum_{m=0}^{x}\sum_{n=0}^{c-1}h(m,n) & \sum_{m=0}^{x}\sum_{n=y+c-H}^{c-1}h(m,n) \\ \sum_{m=0}^{r-1}\sum_{n=0}^{y}h(m,n) & 1 & \sum_{m=0}^{r-1}\sum_{n=y+c-H}^{c-1}h(m,n) \\ \sum_{m=x+r-V}^{r-1}\sum_{n=0}^{y}h(m,n) & \sum_{m=x+r-V}^{r-1}\sum_{n=0}^{c-1}h(m,n) & \sum_{m=x+r-V}^{r-1}\sum_{n=y+c-H}^{c-1}h(m,n) \end{bmatrix} \tag{17}$$

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TABLE II
MAIN CHARACTERISTICS OF THE EXPERIMENTAL DATA

Image ID	Location	Resolution (m)	Size (px)	Bands used	TDI CCD	
					Integration stages	Single integration time (ms)
A	120.1°E, 36.9°N	8	5064 × 1536	B1	24	1.125673453013
				B2	16	
				B3	8	
B	119.2°E, 33.6°N	8	5064 × 1536	B1	24	1.123994588852
				B2	16	
				B3	8	

images, the parallax disparities averaged by line are displayed in Fig. 10. Fig. 10(a) and (b) shows the parallax disparities of image A respectively related to Fig. 9(a) and (b) with B1-B2 and B2-B3 bands used separately. Fig. 10(c) and (d) shows the parallax disparities of image B respectively related to Fig. 9(c) and (d) with B1-B2 and B2-B3 bands used separately. An obvious sinusoidal jitter exits in the across-track direction of the GaoFen-1 02 satellite, but it is not visible in its along-track direction. One explanation for this phenomenon is that the pitch angle is more stable than the roll angle for providing stability to DEM generation, so the satellite jitter in the pitch angle corresponding to the along-track direction is not apparent [11], [17]. In addition, the distortion is significantly affected by the terrain which interferes with the distribution of jitter offsets in the along-track direction [11], [13], [17], [19]. Hence, we only considered the across-track direction. For one set of images, parallax disparities obtained by B1-B2 bands vary from that obtained by B2 and B3 bands. For instance, the amplitude of parallax disparities obtained from B1-B2 bands (around 0.7 pixels) is larger than that from B2 and B3 bands (approximately 0.6 pixels), whereas the frequencies are similar, which is due to the multispectral camera structure of the GaoFen-1 02 satellite [11]. Fig. 11 shows the relationship between camera focal plane, image bands, and parallax images. On the camera focal plane of GaoFen-1 02 satellite, the positions of B1 and B2 bands are 76 pixels apart, while that of B2 and B3 bands are 64 pixels apart, as shown in Fig. 11(a). Fig. 11(b) displays the way the bands arranged due to the placement of the camera focal plane. Fig. 11(c) shows the parallax images generated from the bands. The parallax images are obtained through the overlap of adjacent bands. Different bands are imaged at the same time, and jitter offsets of the overlap vary from band to band; hence, there are differences in parallax disparities between different bands.

Table III shows the fitted parameters of the jitter curve (i.e., frequency, amplitude, and phase) based on the ITF and CDSM in the presented method. The frequencies of jitter curves are about 1.1 Hz, and the amplitudes are around 1.12 pixels and 1.07 pixels, respectively. It is worth noticing that the deviation between the jitter curve parameters respectively estimated from the B1-B2 and B2-B3 bands is very small. The deviation in frequency is no more than 0.0002 Hz, the deviation in amplitude is less than 0.02 pixels, and the deviation in phase is less than 0.003 rad. These indicate that the presented method has the reliability of jitter detection.

TABLE III
PARAMETERS OF THE FITTED JITTER CURVE

Image ID	Bands used	Frequency (Hz)	Amplitude (px)	Phase (rad)
A	B1 and B2	1.1005	1.1156	-3.1254
	B2 and B3	1.1006	1.1252	-3.1228
	deviation	0.0001	0.0096	0.0026
B	B1 and B2	1.1000	1.0613	-0.5927
	B2 and B3	1.1002	1.0734	-0.5942
	deviation	0.0002	0.0121	0.0015

Due to their high geometric accuracy and ability to reflect the real ground conditions, the 2-m orthoimages are further utilized to obtain the reference value of the jitter curve. For objectivity, ERDAS IMAGINE is used for image registration between each band of image A/B and their corresponding orthoimages through dense ground control point (GCP) matching. A total of 2447, 2698, and 2626 GCPs are obtained for image A B1, B2, and B3 bands, respectively, and 2385, 2465, and 2253 GCPs are obtained for image B B1, B2, and B3 bands, respectively. The registration residuals between image bands and orthoimages are shown in Fig. 12. Different from that the calculation of parallax images contains $\varepsilon(t)$-to-$J(i)$ evolving process as described by the ITF, the calculation of registration residuals between bands and orthoimages only involves $\varepsilon(t)$-to-$E(i)$ conversion, which represents the integration imaging of TDI CCD. According to (1) (Section II-A), these residuals ($E(i)$) can be estimated as follows:

$$\begin{aligned} E(i) &= \frac{1}{N}\left[\frac{1}{2}\varepsilon((i-N)T)+\sum_{k=1}^{N-1}\varepsilon((i-N+k)T)+\frac{1}{2}\varepsilon(iT)\right] \\ &= \frac{1}{N}\left[\frac{1}{2}A\sin\left(\frac{2\pi Ti}{T_j}-\frac{2\pi TN}{T_j}+\theta\right)\right. \\ &\quad +\sum_{k=1}^{N-1}A\sin\left(\frac{2\pi Ti}{T_j}+\frac{2\pi T}{T_j}(k-N)+\theta\right) \\ &\quad \left.+\frac{1}{2}A\sin\left(\frac{2\pi Ti}{T_j}+\theta\right)\right] \end{aligned} \tag{18}$$

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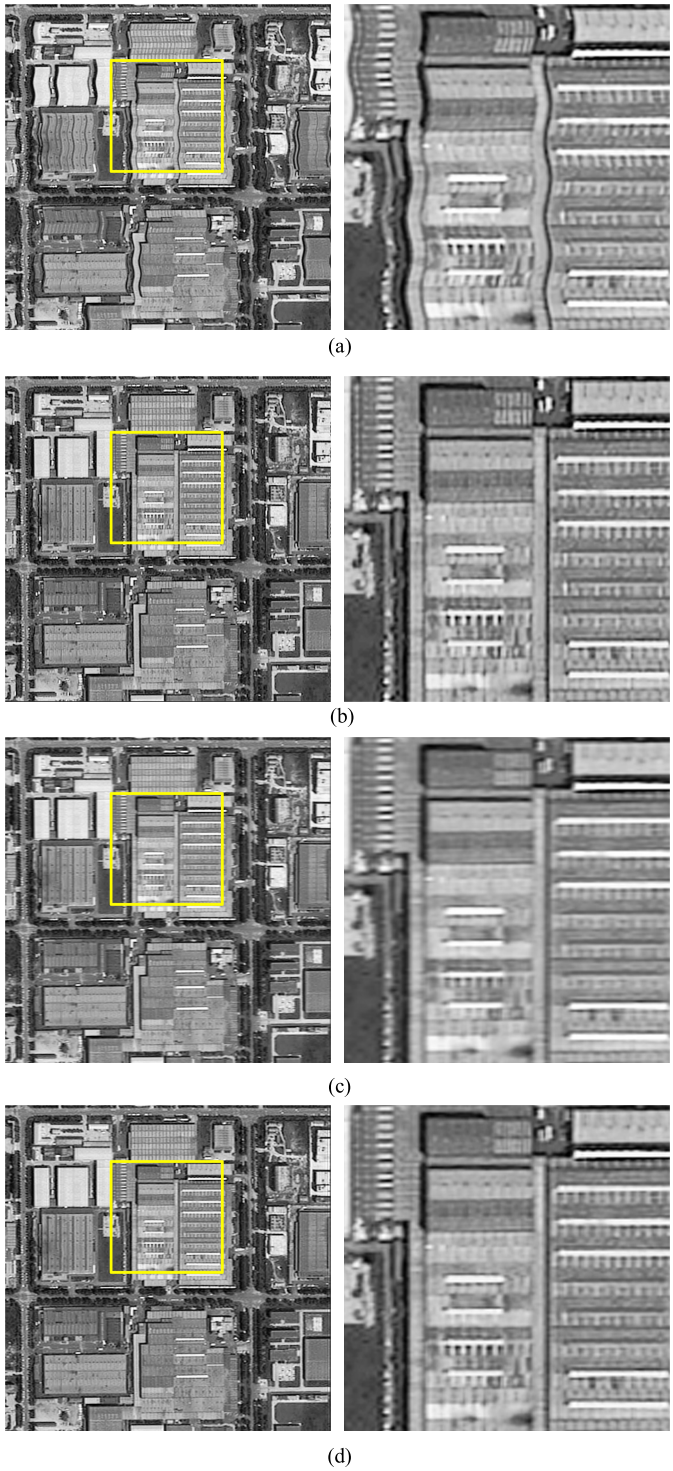

Fig. 7. (Left–Right) Results of simulation experiments and details of the local region. (a) Simulated image considering 0.1 time ratio jitter. (b) Restored image obtained from (a). (c) Simulated image considering a 2.0 time ratio jitter. (d) Restored image obtained from (c).

where i is the line number, N is the TDI CCD integration stages, T is the time needed for one integration, $\varepsilon(t)$ is the jitter curve and can be described by the sine function, in which A is the amplitude, T_j is the period of the jitter curve, and θ is the initial phase.

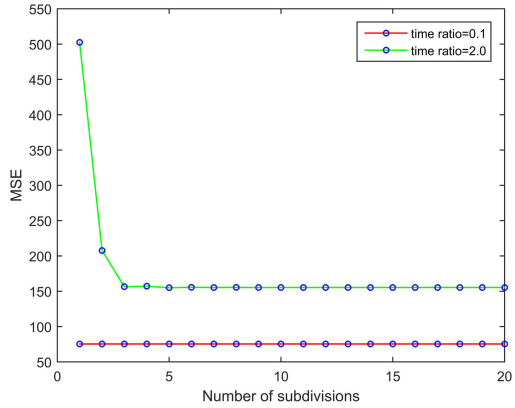

Fig. 8. Relationship between MSE and number of subdivisions in the CDSM.

TABLE IV
REFERENCE VALUE OF THE JITTER CURVE

Image ID	Bands used	Frequency (Hz)	Amplitude (px)	Phase (rad)
A	B1	1.1003	1.1018	-3.1130
	B2	1.1009	1.1084	-3.1290
	B3	1.1006	1.1011	-3.1256
	mean	1.1006	1.1038	-3.1225
B	B1	1.1009	1.0404	-0.6206
	B2	1.1009	1.0445	-0.6222
	B3	1.1002	1.0424	-0.5905
	mean	1.1007	1.0424	-0.6111

On the basis of (18), the reference value of the jitter curve can be derived from the registration residuals between image bands and orthoimages. The specific parameters of the jitter curve are provided in Table IV. Because of many inevitable errors such as fitting error, the parameters of the jitter curve calculated by different bands are slightly different, whose mean values are recorded in Table IV. According to Tables III and IV, the accuracy of the jitter detection within the presented method can be evaluated. In order to further illustrate the performance of the presented method, a comparison with the method based on the jitter displacement estimation model [20] is conducted, and the results are shown in Table V. With the presented method used, the difference between detected frequency and the reference value is no more than 0.0007 Hz, the error of the detected amplitude does not exceed 0.031 pixels, and the error of the detected phase is less than 0.02 rad. Meanwhile, the error of parameters of the jitter curve detected by the presented method is generally smaller than that obtained by the method based on the jitter displacement estimation model. The comparison proofed the advantages of the presented method, which benefits from the detailed analysis of the imaging process with ITF and CDSM used. The presented method connects the image space with the real object space, and the overall effect with immediate effects by ITF, whereas the focus of the method based on the jitter displacement estimation model is only on the image space,

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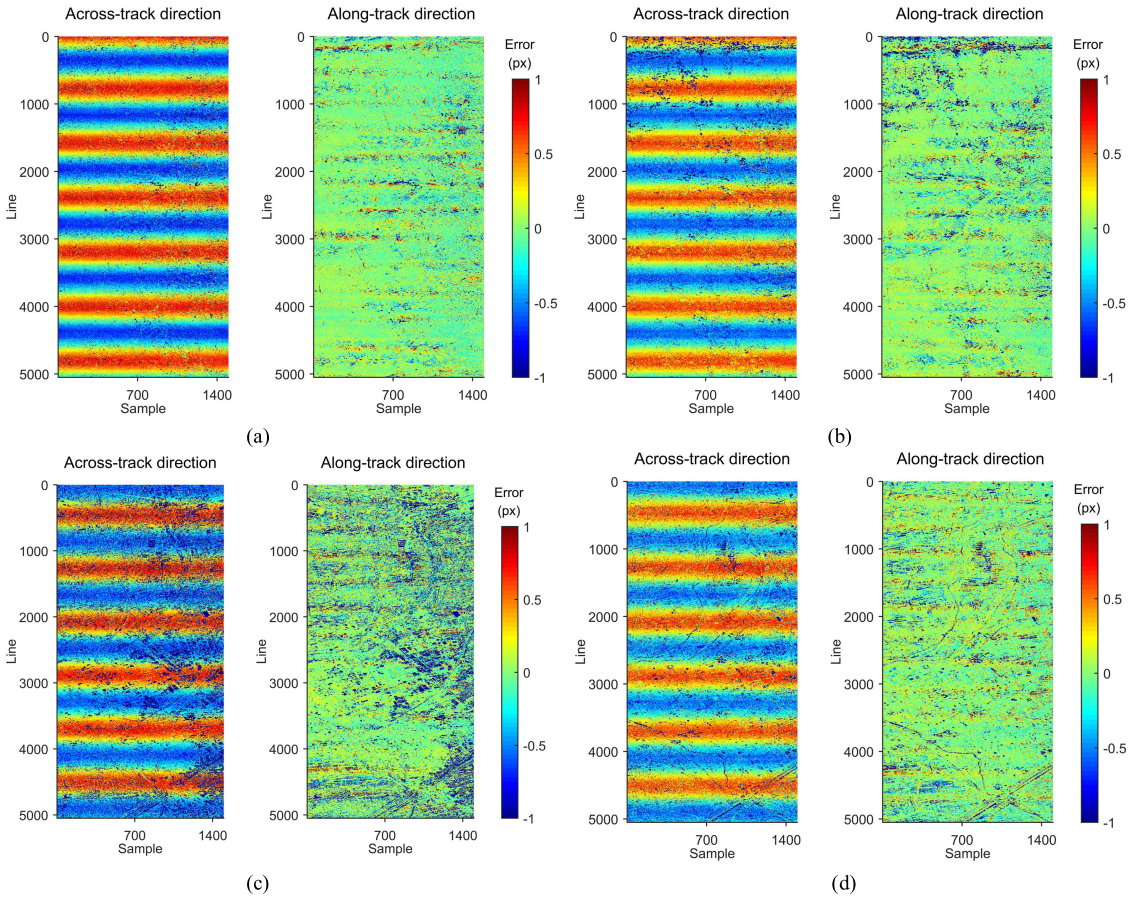

Fig. 9. Obtained parallax images after subpixel matching. (a) Parallax images of image A, using B1 and B2 bands for matching. (b) Parallax images of image A, using B2 and B3 bands for matching. (c) Parallax images of image B, using B1 and B2 bands for matching. (d) Parallax images of image B, using B2 and B3 bands for matching.

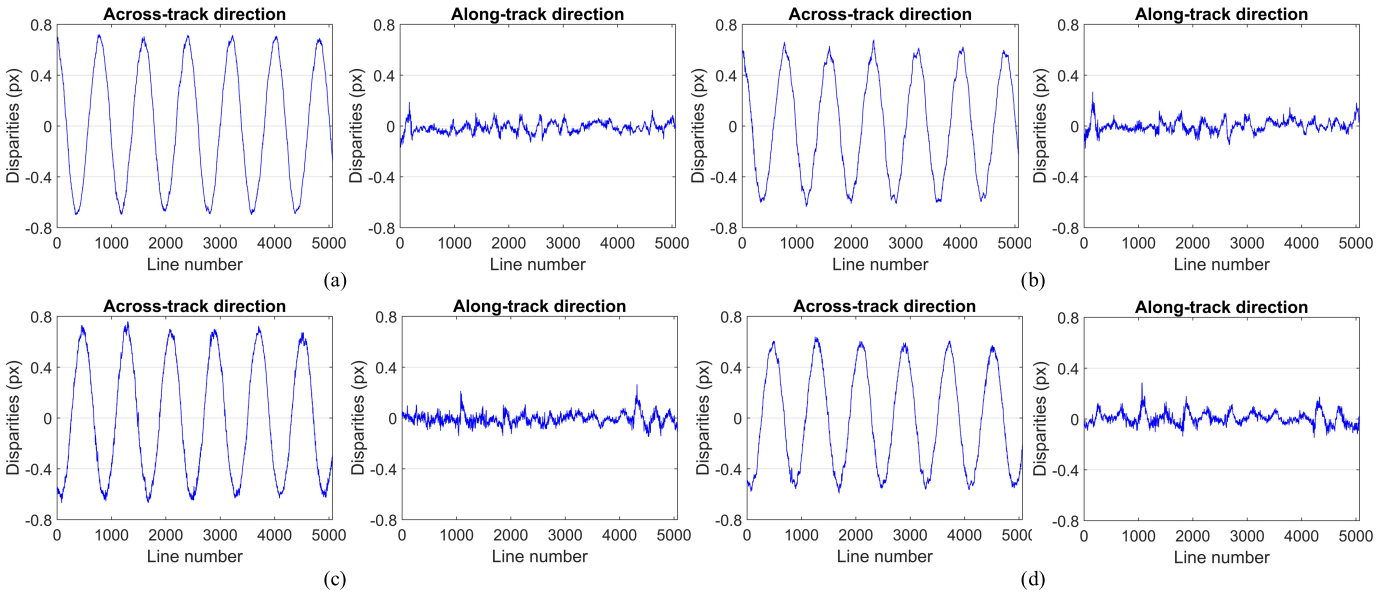

Fig. 10. Parallax disparities. (a) Parallax disparities of image A, with B1 and B2 bands used. (b) Parallax disparities of image A, with B2 and B3 bands used. (c) Parallax disparities of image B, with B1 and B2 bands used. (d) Parallax disparities of image B, with B2 and B3 bands used.

which cannot reflect the detailed change in jitter. Besides this, the presented method adopted CDSM to reflect the continue push-broom mode, while the method based on the jitter displacement estimation model cannot do that with the

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TABLE V
DETECTED PARAMETERS OF THE JITTER CURVE

	Image A								Image B							
	The method based on the jitter displacement estimation model				The presented method				The method based on the jitter displacement estimation model				The presented method			
	B1 and B2		B2 and B3		B1 and B2		B2 and B3		B1 and B2		B2 and B3		B1 and B2		B2 and B3	
	detected	error	detected	error	detected	error	detected	error	detected	error	detected	error	detected	error	detected	error
Frequency (Hz)	1.1005	0.0001	1.1006	0	1.1005	0.0001	1.1006	0	1.1000	0.0007	1.1002	0.0005	1.1000	0.0007	1.1002	0.0005
Amplitude (px)	1.1713	0.0675	1.1934	0.0896	1.1156	0.0118	1.1252	0.0214	1.1142	0.0718	1.1385	0.0961	1.0613	0.0189	1.0734	0.0310
Phase (rad)	-3.2045	0.0820	-3.1704	0.0479	-3.1254	0.0029	-3.1228	0.0003	-0.6716	0.0605	-0.6417	0.0306	-0.5927	0.0184	-0.5942	0.0169

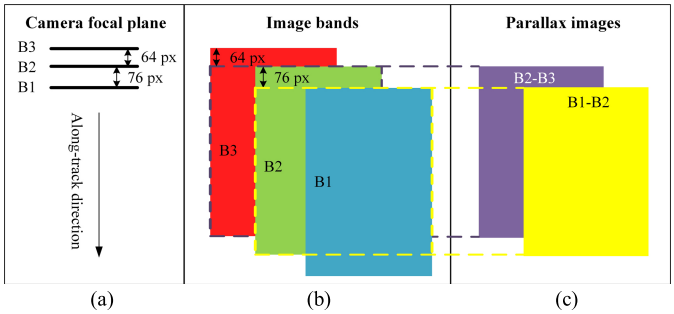

Fig. 11. Imaging information of GaoFen-1 02 satellite. (a) Multispectral camera focal plane (showing that installed with the three adopted bands). (b) Way the bands are arranged. (c) Parallax images.

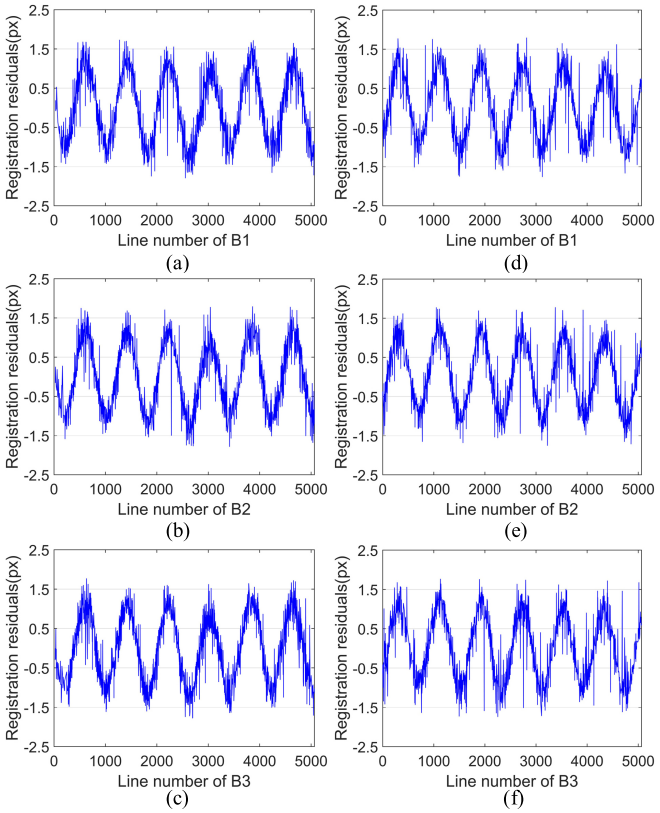

Fig. 12. Registration residuals between image bands and orthoimages. (a) Residuals of the B1 band of image A. (b) Residuals of the B2 band of image A. (c) Residuals of the B3 band of image A. (d) Residuals of the B1 band of image B. (e) Residuals of the B2 band of image B. (f) Residuals of the B3 band of image B.

traditional sampling strategy used. Thus, the presented method shows better performance on jitter detection.

b) Results of image restoration: One note about the experiments is that images of each band were restored based on the detected jitter curve, then the distance between the used bands was eliminated by translation, and the true color image was synthesized for display.

Experiments were conducted through a combination of methods, respectively, in two stages, i.e., jitter detection and image restoration, to research on the restoration issue and explore the influence of jitter detection on image restoration at the same time. For the comparison in the jitter detection stage, the results obtained by the method based on the jitter displacement estimation model in the part "results of jitter detection" were used. As for the comparison in image restoration, a stepwise restoration method was adopted, where Tong *et al.*'s [8] jitter compensation on the image method was employed first for eliminating distortion and then the blind image restoration based on the maximum likelihood algorithm [42] was applied for deblurring. In another word, geometric quality and radiometric quality were taken into account simultaneously in the presented method, whereas the stepwise restoration method considered them separately for comparison.

The restoration performances of the experiments are demonstrated in Figs. 13 and 14. For image A, the original image is shown in Fig. 13(a); the details of region I in Fig. 13(a)

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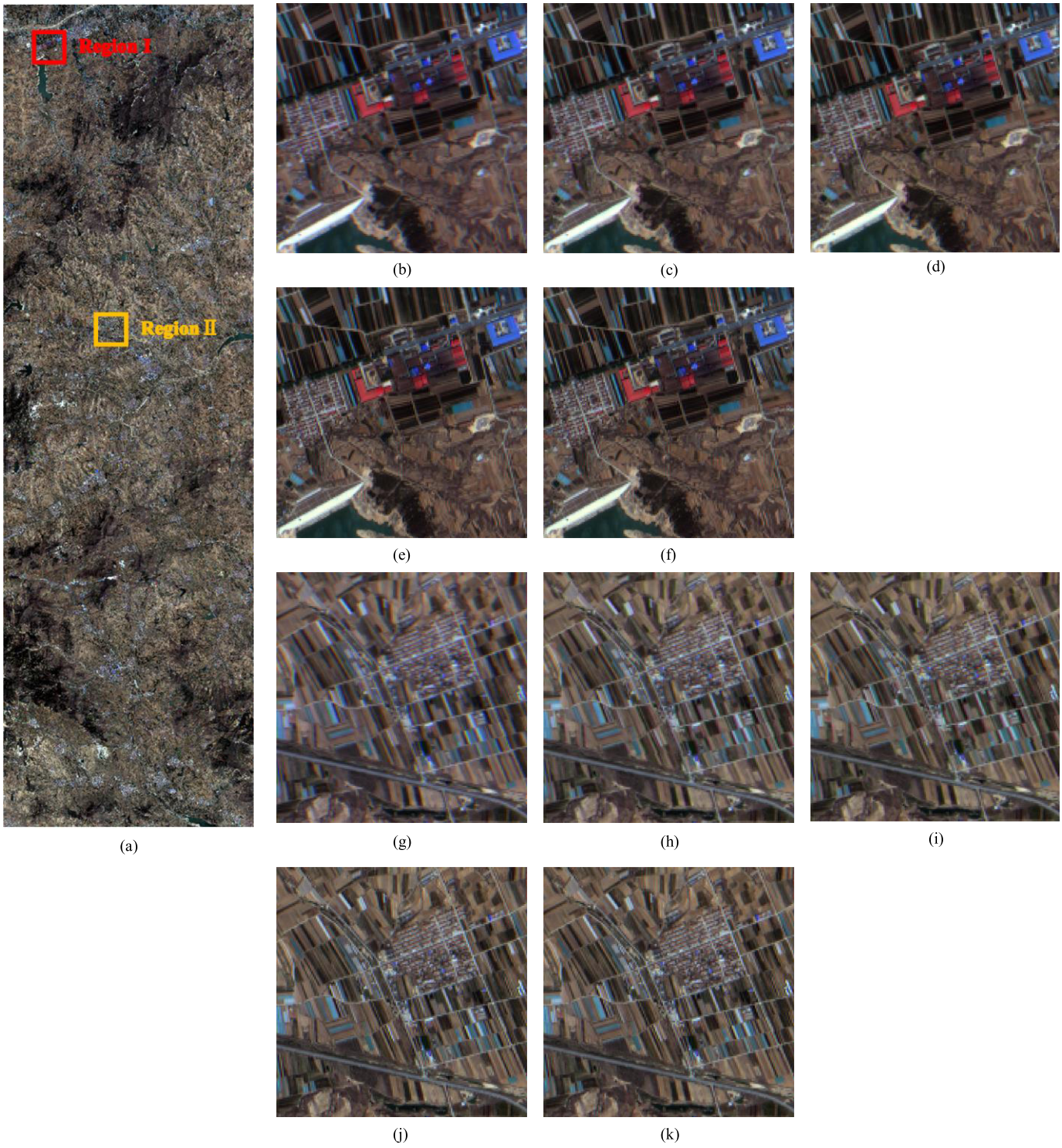

Fig. 13. Experimental results of image A. (a) Original image. (b) Details of region I in (a). (c) Result of region I using Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method. (d) Result of region I using Detected by the presented method + Restored by the stepwise restoration method. (e) Result of region I using Detected by the method based on the jitter displacement estimation model + Restore by the presented method. (f) Result of region I using Detected by the presented method + Restore by the presented method. (g) Details of region II in (a). (h) Result of region II using Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method. (i) Result of region II using Detected by the presented method + Restored by the stepwise restoration method. (j) Result of region II using Detected by the method based on the jitter displacement estimation model + Restore by the presented method. (k) Result of region II using Detected by the presented method + Restore by the presented method.

are shown in Fig. 13(b); the obtained results of region I are displayed in Fig. 13(c)–(f), respectively. Fig. 13(c) shows the result of region I using Detected by the method based on the jitter displacement estimation model + Restored by the

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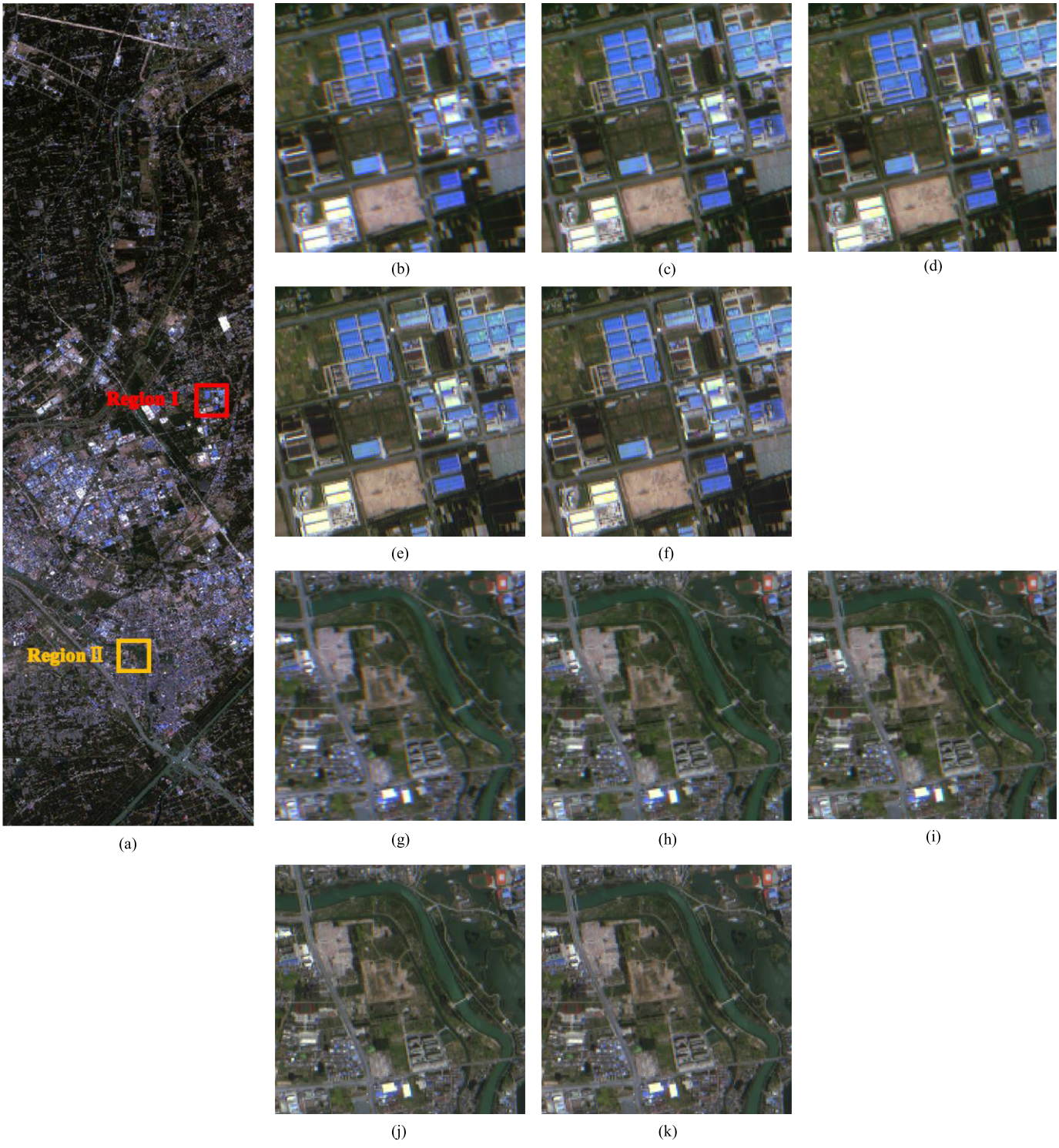

Fig. 14. Experimental results of image B. (a) Original image. (b) Details of region I in (a). (c) Result of region I using Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method. (d) Result of region I using Detected by the presented method + Restored by the stepwise restoration method. (e) Result of region I using Detected by the method based on the jitter displacement estimation model + Restore by the presented method. (f) Result of region I using Detected by the presented method + Restore by the presented method. (g) Details of region II in (a). (h) Result of region II using Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method. (i) Result of region II using Detected by the presented method + Restored by the stepwise restoration method. (j) Result of region II using Detected by the method based on the jitter displacement estimation model + Restore by the presented method. (k) Result of region II using Detected by the presented method + Restore by the presented method.

stepwise restoration method ("+" represents followed by). Fig. 13(d) shows the result of region I using Detected by the presented method + Restored by the stepwise restoration method. Fig. 13(e) shows the result of region I using Detected

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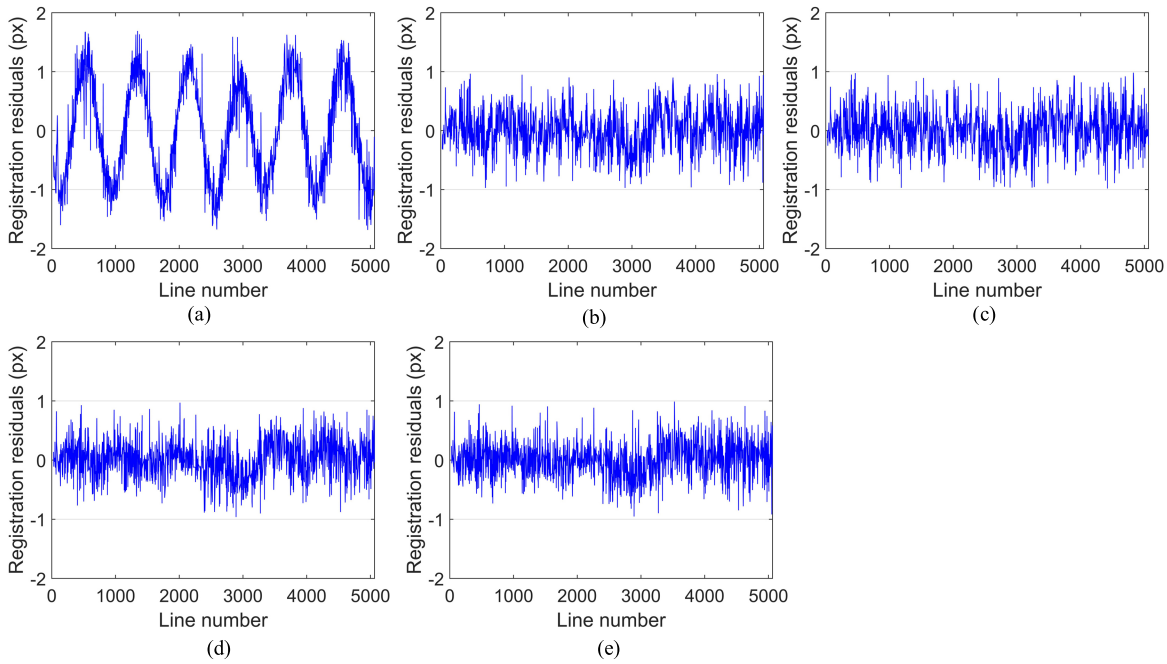

Fig. 15. Registration residuals of image A. (a) For Original image. (b) For obtained image using Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method. (c) For obtained image using Detected by the presented method + Restored by the stepwise restoration method. (d) For obtained image using Detected by the method based on the jitter displacement estimation model + Restore by the presented method. (e) For obtained image using Detected by the presented method + Restore by the presented method.

by the method based on the jitter displacement estimation model + Restore by the presented method. Fig. 13(f) shows the result of region I using Detected by the presented method + Restore by the presented method. The details of region II in Fig. 13(a) are shown in Fig. 13(g); the obtained results of region II are displayed in Fig. 13(h)–(k), respectively. Fig. 13(h) shows the result of region II using Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method, Fig. 13(i) shows the result of region I using Detected by the presented method + Restored by the stepwise restoration method, Fig. 13(j) shows the result of region II using Detected by the method based on the jitter displacement estimation model + Restore by the presented method, Fig. 13(k) shows the result of region II using Detected by the presented method + Restore by the presented method, Similarly, the results of image B are shown in Fig. 14.

It is obvious that a visible blur exists in original images, while the clarity of result images is improved to varying degrees. As to the visual effect, Fig. 14(c) and (d), (e) and (f), (h) and (i), and (j) and (k) are the similar pairs, in which the difference almost indistinguishable by eyes. The approaches to obtain result images in a similar pair differ only in jitter detection (i.e., one uses the method based on the jitter displacement estimation model, and the other adopts the presented method). According to Table V, the difference between values detected by both methods is very small. Hence, it is reasonable for similar pairs to have a close performance. Nevertheless, the restored images obtained by the presented method have better visual effect than those obtained by the stepwise restoration method, which can be seen from the comparison between the pair consisting of Fig. 14(c) and (d) and the pair consisting of Fig. 14(e) and (f), or from the comparison between the pair consisting of Fig. 14(h) and (i) and the pair consisting of Fig. 14(j) and (k).

In addition to the visualized results, Figs. 15 and 16 show the registration residuals between images A/B with their related 2-m orthoimages. For image A, a total of 1765 GCPs corresponding to the original image were obtained as shown in Fig. 15(a); 1782 GCPs, corresponding to the resulting image obtained using Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method, were obtained as shown in Fig. 15(b); 1782 GCPs, corresponding to the resulting image obtained using Detected by the presented method + Restored by the stepwise restoration method, were obtained as shown in Fig. 15(c); 1620 GCPs, corresponding to the resulting image obtained using Detected by the method based on the jitter displacement estimation model + Restored by the presented method, were obtained as shown in Fig. 15(d); and 1568 GCPs, corresponding to the resulting image obtained using Detected by the presented method + Restore by the presented method, were obtained as shown in Fig. 15(e). For image B, a total of 1685 GCPs corresponding to the original image were obtained as shown in Fig. 16(a); 1703, 1685, 1548, and 1443 GCPs were obtained corresponding to the restored images, respectively, obtained using Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method, Detected by the presented method + Restored by the stepwise restoration method, Detected by the method based on the jitter displacement estimation model + Restore by the presented

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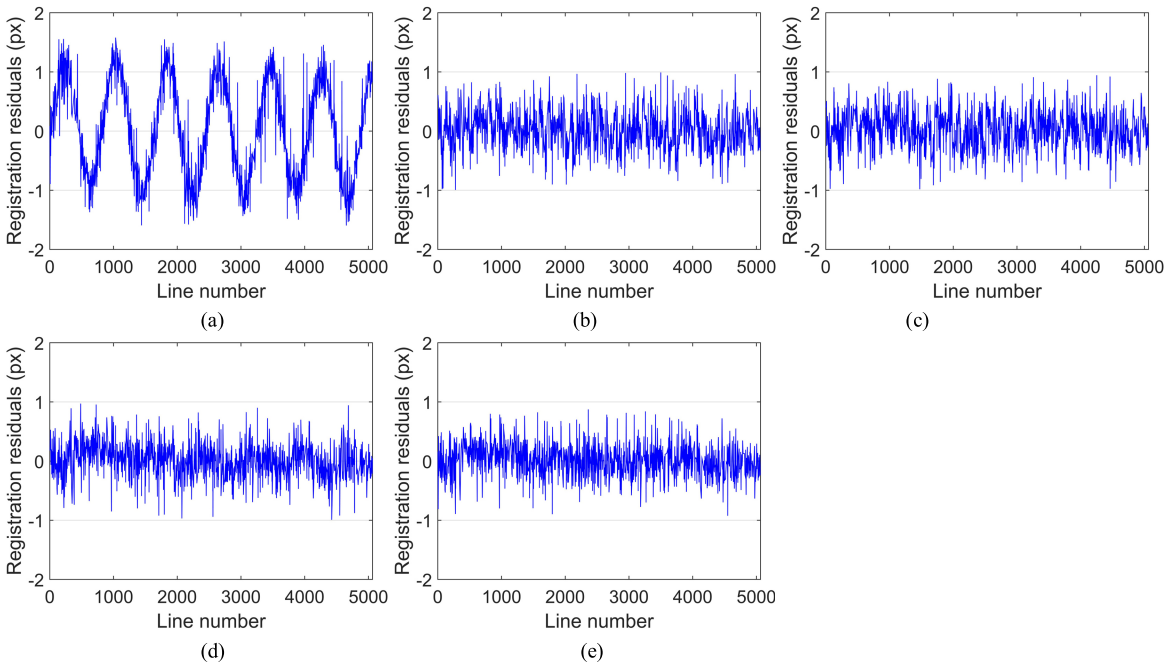

Fig. 16. Registration residuals of image B. (a) For Original image. (b) For obtained image using Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method. (c) For obtained image using Detected by the presented method + Restored by the stepwise restoration method. (d) For obtained image using Detected by the method based on the jitter displacement estimation model + Restore by the presented method. (e) For obtained image using Detected by the presented method + Restore by the presented method.

method, and Detected by the presented method + Restore by the presented method. It is worth noting that the sinusoidal offsets obviously occur in original images, while they are compensated in the restored images. Since the registration residuals of restored images obtained by different methods in Figs. 15 and 16 looked similar, quantitative comparisons were further conducted to evaluate the geometric and radiometric performances.

Details of image quality evaluation before and after restoration were shown in Table VI. In the geometric performance, mean absolute error (MAE) and root MSE (RMSE) were adopted as the evaluation indexes. According to Table VI, MAE and RMSE values, which are calculated from the images restored by the presented method, are slightly smaller than those calculated from the images obtained after the same method used for jitter detection; however, the stepwise method is used for image restoration. It indicates that the presented method is slightly better than the stepwise restoration method in the geometric aspects. In the radiometric performance, contrast, average gradient, correlation, and Roberts' gradient [28], [36], which were calculated based on the pixels of the whole image, were adopted as the evaluation indexes as shown in Table VI. Among them, using a gray-level co-occurrence matrix (GLCM), correlation reflected the correlation of local gray values in the image. If the image is blurred, its correlation will be large. According to Table VI, all the radiometric evaluations show that, in the case of the same restoration method, the restored result obtained by using the presented method in the jitter detection stage is better than that obtained using the method based on the jitter displacement estimation model. This is because the jitter detection accuracy of the presented method is higher than that of the method based on the jitter displacement estimation model. When the restoration method is the same, higher precision jitter detection will lead to a better image restoration effect. As for the restoration, the presented method shows better performance than the stepwise restoration method, when the used jitter detection methods are the same. Take the contrast for example, in the case where the presented method was adopted in jitter detection, the contrast increased from 17 308.4169 to 21 232.1616 (image A) and from 19 254.7527 to 24 193.4485 (image B) when the stepwise restoration method is replaced by the presented method. The radiometric quality of images is improved significantly. In summary, the presented method is slightly better than the stepwise restoration method in the geometric aspect, while it has advantages in the radiometric aspect. This is mainly because the stepwise restoration method introduces the radiometric error to appear when it is first used to eliminate the jitter distortion. The presented method, however, treats distortion and blur as a coupled effect, and simultaneously deals with the geometric and radiometric problem of images, therefore there is no such additional problems. Besides this, the method proposed in this article is mainly dependent on the presented ITF and CDSM. ITF decomposes the overall effect in immediate effects and extracts the immediate jitter offsets in the object space. CDSM provides a dense sampling strategy to take fully into account the jitter temporal variation. Then, the continue push-broom mode of TDI CCD can be reflected more accurately. In addition, the estimated PSF is also time-varying and space-varying. Therefore, the distortion and blur in high-resolution TDI CCD images affected by satellite jitter can be compensated simultaneously in restoration. Since temporal

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TABLE VI
IMAGE QUALITY EVALUATION BEFORE AND AFTER RESTORATION

		Geometric evaluation		Radiometric evaluation			
		MAE (px)	RMSE (px)	Contrast	Average gradient	Correlation	Roberts gradient
Image A	Untreated	0.7229	0.8229	14704.9812	128.8486	0.8928	161.9966
	Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method	0.2726	0.3529	17172.5298	133.1205	0.8780	167.0169
	Detected by the presented method + Restored by the stepwise restoration method	0.2703	0.3526	17308.4169	133.1737	0.8769	167.0877
	Detected by the method based on the jitter displacement estimation model + **Restore by the presented method**	0.2442	0.3191	21195.9949	147.5444	0.8418	179.4664
	Detected by the presented method + **Restore by the presented method**	0.2151	0.2678	21232.1616	147.6397	0.8416	179.5811
Image B	Untreated	0.6832	0.7819	16682.8921	129.6487	0.9372	165.4379
	Detected by the method based on the jitter displacement estimation model + Restored by the stepwise restoration method	0.2555	0.3364	19010.4243	133.7642	0.9293	170.2548
	Detected by the presented method + Restored by the stepwise restoration method	0.2540	0.3308	19254.7527	133.8396	0.9283	170.3533
	Detected by the method based on the jitter displacement estimation model + **Restore by the presented method**	0.2092	0.2741	24155.6475	146.6572	0.9049	181.5550
	Detected by the presented method + **Restore by the presented method**	0.1904	0.2416	24193.4485	146.7722	0.9048	181.6253

variation, spatial distribution, and spectrum characteristics are coupled together, the presented method can obtain better geometric and radiometric performance.

IV. CONCLUSION

In this article, we presented a method containing jitter detection and image restoration for high-resolution TDI CCD satellite images. The main innovation of this method lies in the proposed ITF and CDSM. The ITF aims at improving jitter detection performance. Traditional jitter detection methods generally consider the overall image motion in the image space. In contrast, the proposed method considers the conversion of the jitter from the image space to the object space using ITF, as well as the decomposition of the overall effect in immediate effects (by converting the parallax images into jitter curves). Due to the fact that ITF can extract the immediate jitter offsets in the object space, it shows better performance compared with traditional jitter detection methods. The CDSM is built to fully consider the jitter temporal variation and the continuous push-broom mode of the TDI CCD. Many previous methods consider the number of exposures of TDI CCD as equal to its number of integration stages. As to CDSM, it provides a dense sampling strategy based on subdivision. Using CDSM, the discrete issue can be approximated highly to the continuous issue, and the continue push-broom mode of TDI CCD also would be reflected more accurately. The CDSM contributes to both jitter detection and image restoration. The reasonable subdivision number of CDSM is determined by theoretical analysis and is also verified in experiments. Meanwhile, the restoration results of the simulated images affected by high- and low-time ratio jitters indicated that the method proposed in this article is suitable for jitters having a wide range of frequencies. Images of GaoFen-1 02 satellite are adopted in real image restoration experiments. Quantitative evaluation and comparison with the stepwise method show the presented method has advantages on jitter detection, as well as improvement of the geometric and radiometric quality of images: it can simultaneously compensate for the distortion and blur effects caused by satellite jitter.

Nevertheless, the subpixel matching based on the correlation coefficient and least squares algorithm used in this article for high-frequency jitter detection may have its limitations.

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Other more accurate matching methods, e.g., the deep learning framework for remote sensing image registration [37], may further increase the accuracy of parallax images so as to improve jitter detection performance. In addition, the presented method used to deal with multispectral images would be migrated to panchromatic images. These issues will be addressed in the future.

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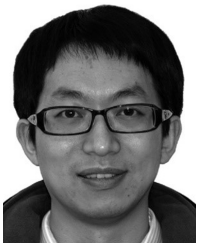

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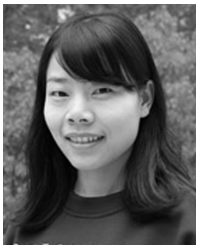

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